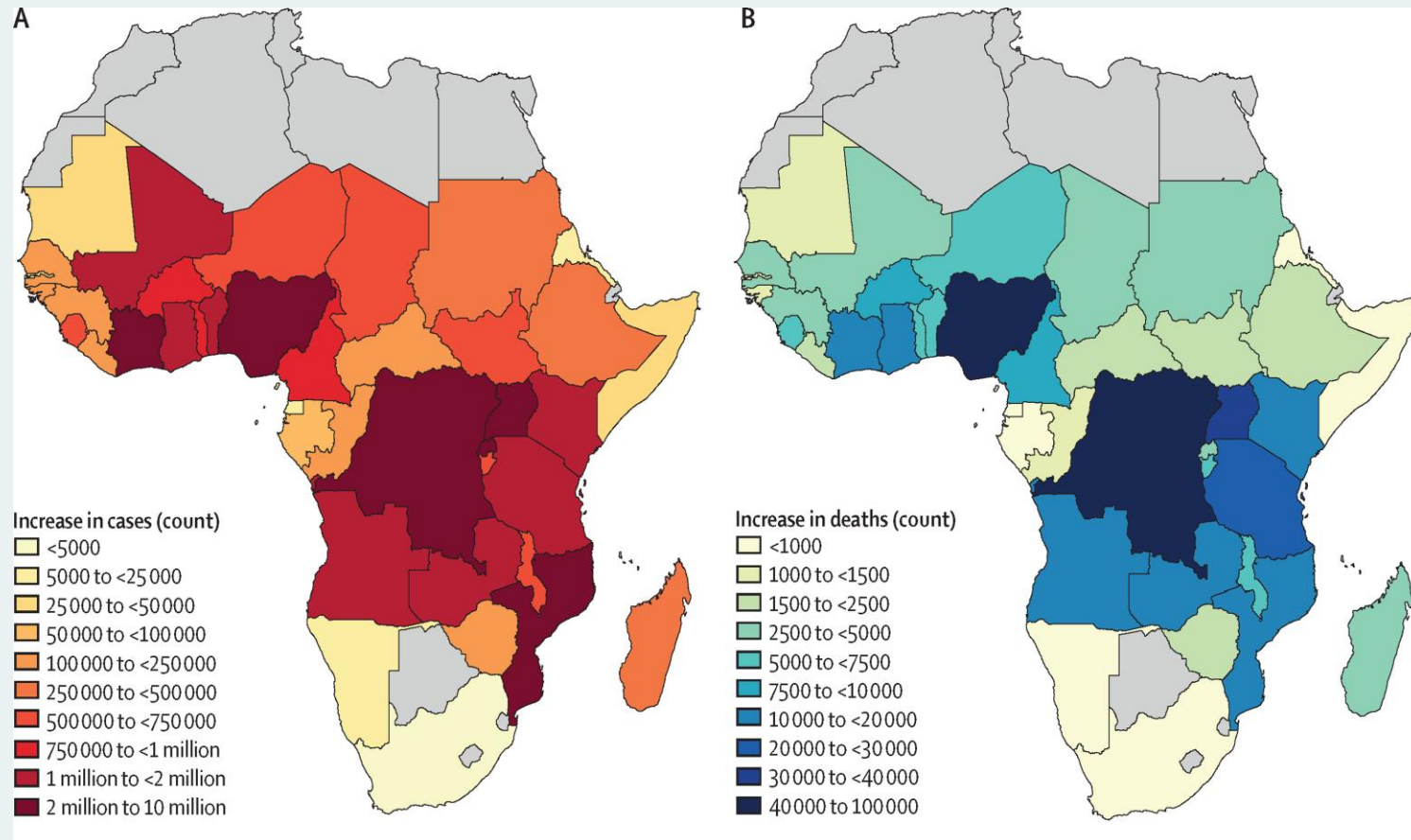


Immune Response Within Malaria Patients

Edem Wordi
and
Jaimie Harnois

Background

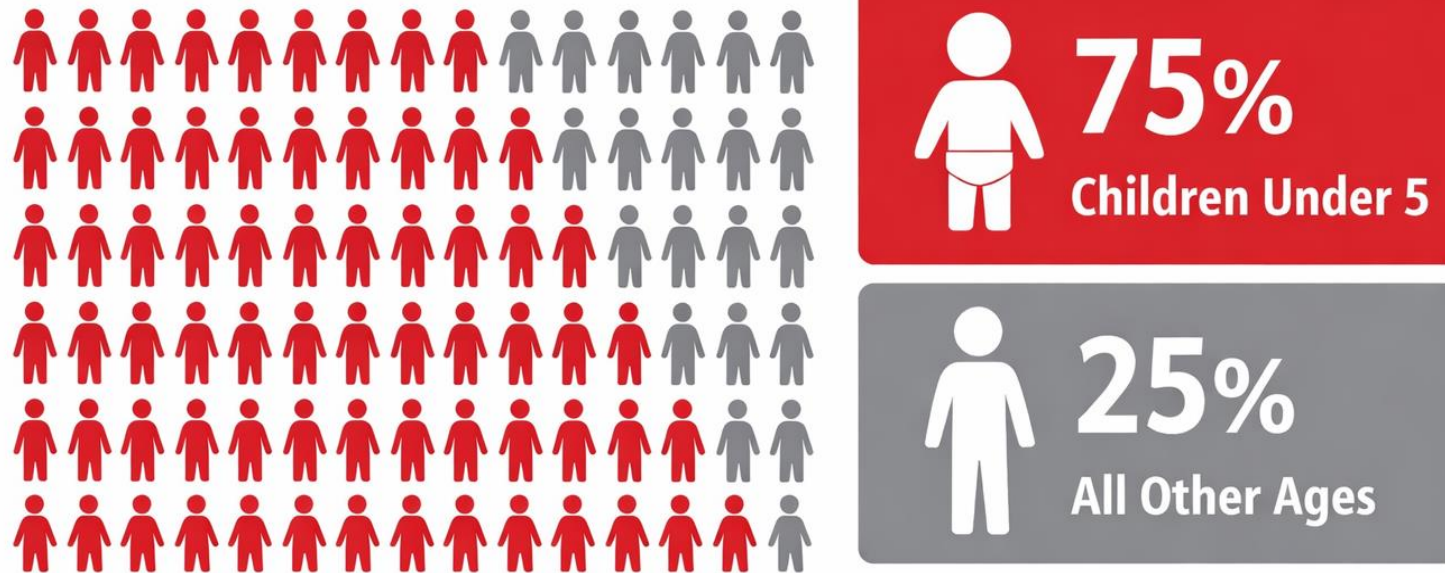


Malaria is the leading cause of death for children in much of Sub-Saharan Africa

**Our World
in Data**

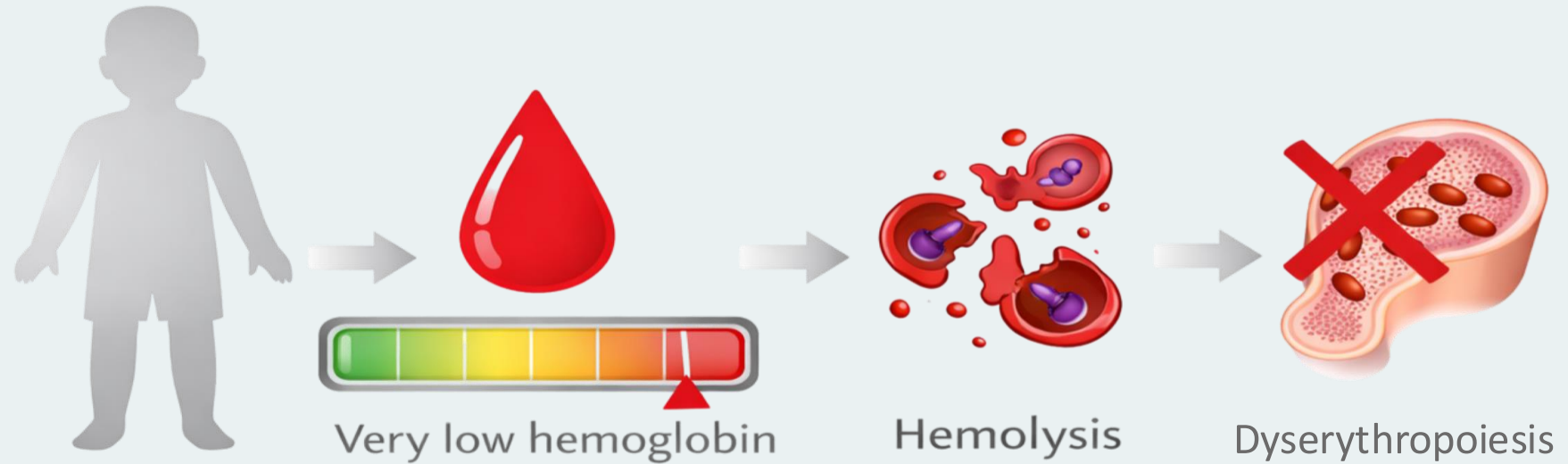
Background

Malaria Deaths in Sub-Saharan Africa



Children under 5 account for about three-quarters of all malaria deaths in the African region.

Background



Severe Malaria Anemia (SMA)

We expect our immune system to protect us. However, malaria can cause our immune system to overwork causing us harm.

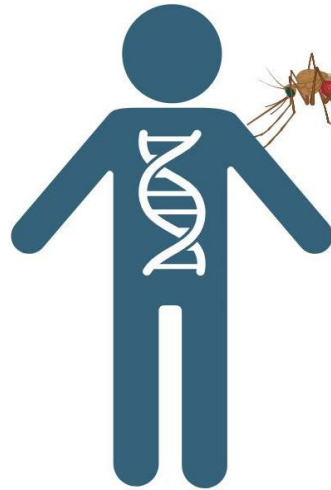
Research Paper Interpretation

“Transcriptomic and Proteomic Insights into Host Immune Responses in Pediatric Severe Malarial Anemia: Dysregulation in HSP60-70-TLR2/4 Signaling and Altered Glutamine Metabolism” (Onyango, C. et al (2024))

- Integrated transcriptomic and proteomic data related to host immune responses to interpret specific molecular mechanisms underlying clinical severities of *P. falciparum* infection.
- Whole-blood RNA sequencing (RNA-seq) of Kenyan pediatric malaria cases. Children categorized by hemoglobin:
 - SMA: Hb < 6.0 g/dL
 - non-SMA: Hb ≥ 6.0 g/dL
- Focused on TLR signaling

Severe Malarial Anemia Pathogenesis: HSP60-HSP70-TLR2/4 Signaling

Child Presents at Hospital with
Acute Febrile Illness



n=577



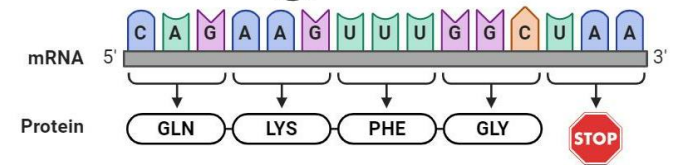
SMA (n=25)



non-SMA (n=41)

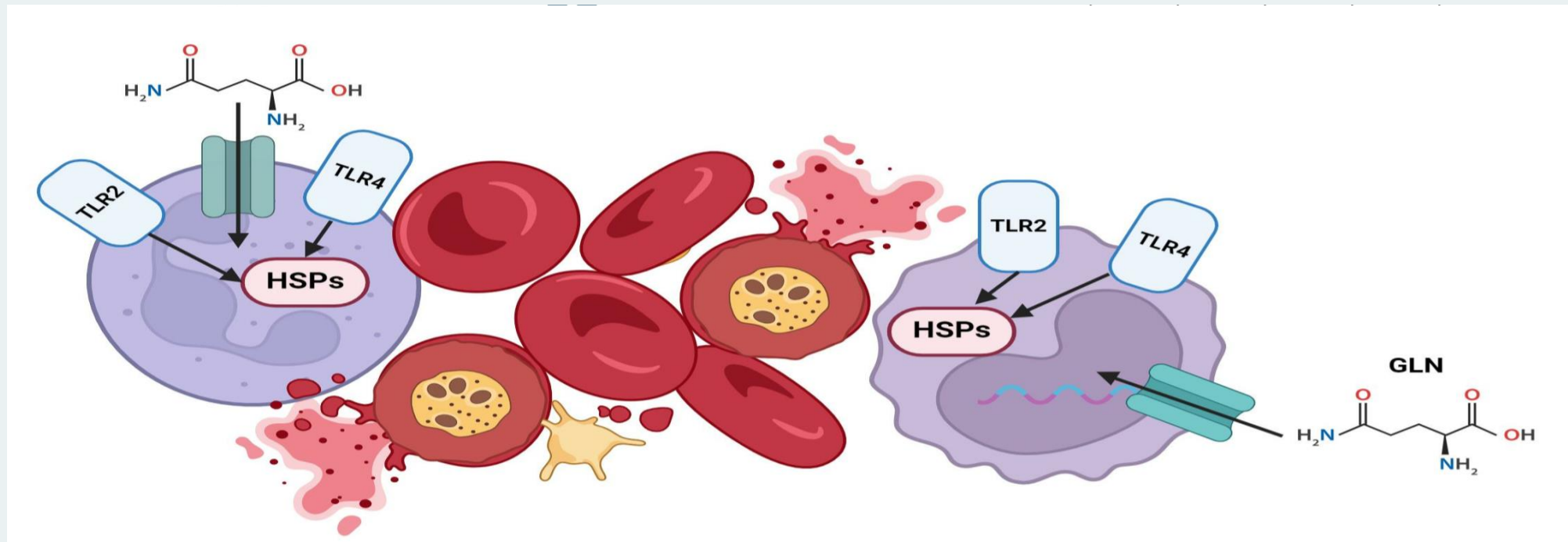


Entire expressed Transcriptome
Proteomics (7,000 plasma proteins)



GLN



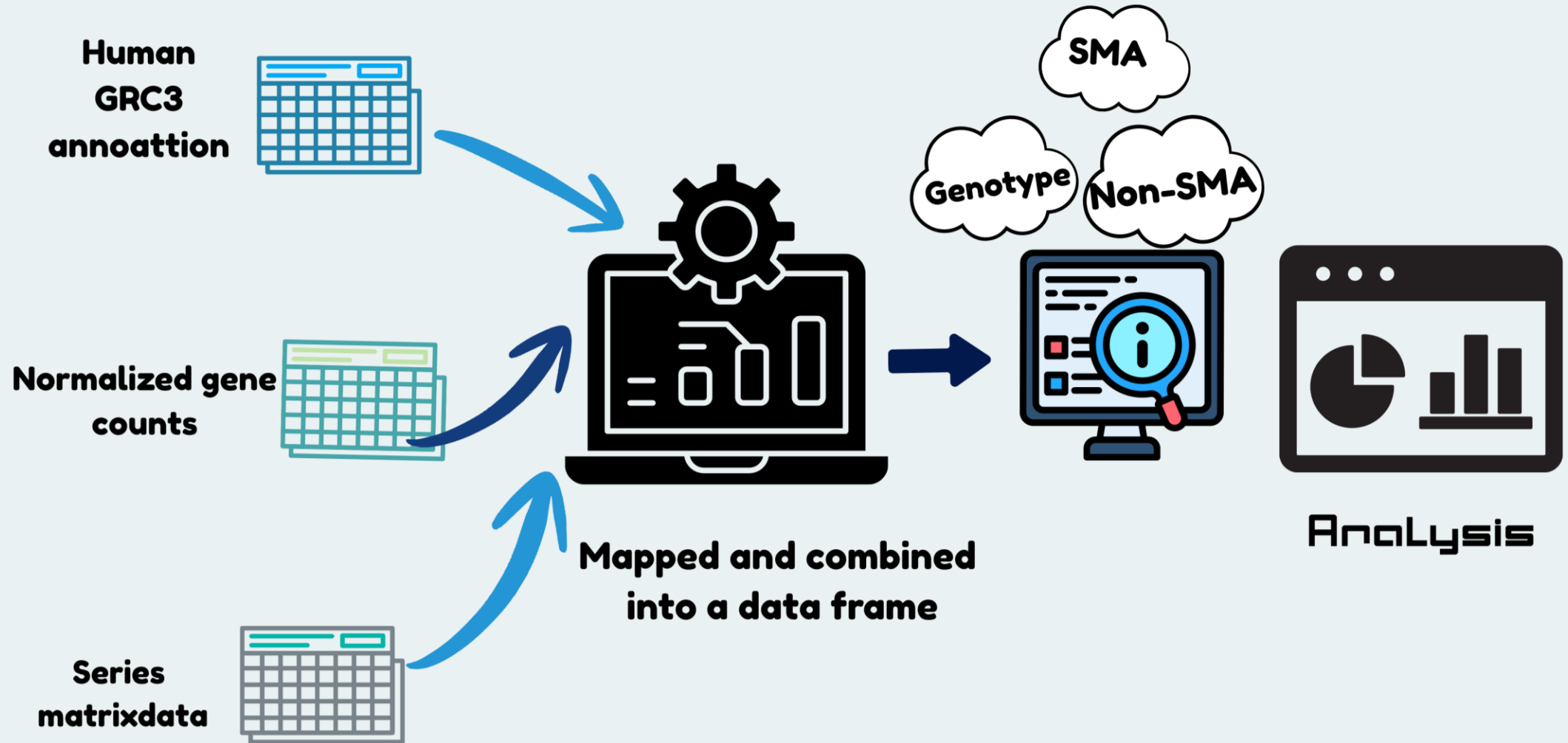


Findings:

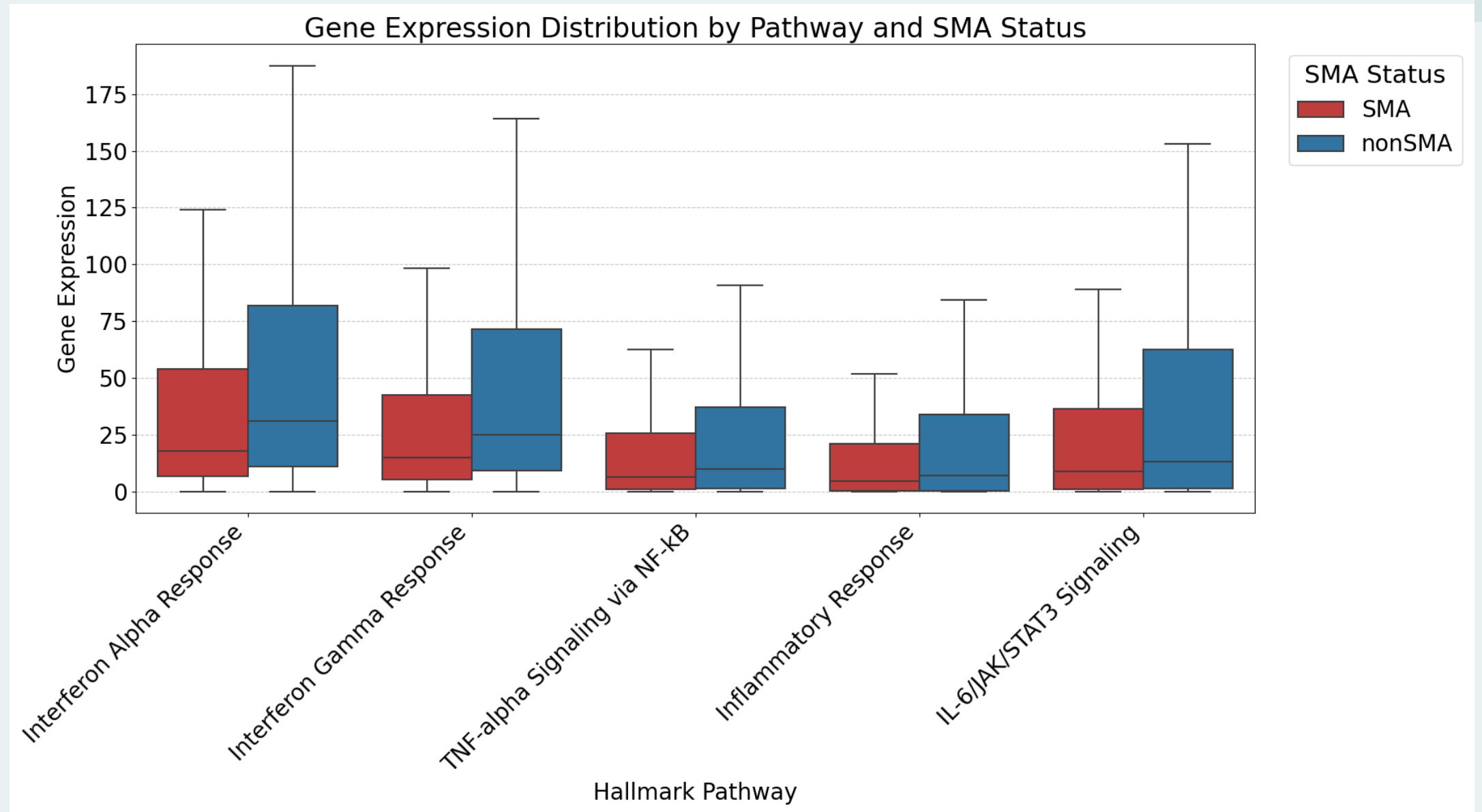
- Distinct gene and protein expression profiles associated with SMA severity.
- Heightened cellular stress response, including heat shock proteins and activation of toll like receptor signaling.
- Excessive immune dysregulation which contributes to anemia severity.

Do immune-related gene expression patterns and pathways differ between children with severe malarial anemia (SMA) compared to children without SMA?

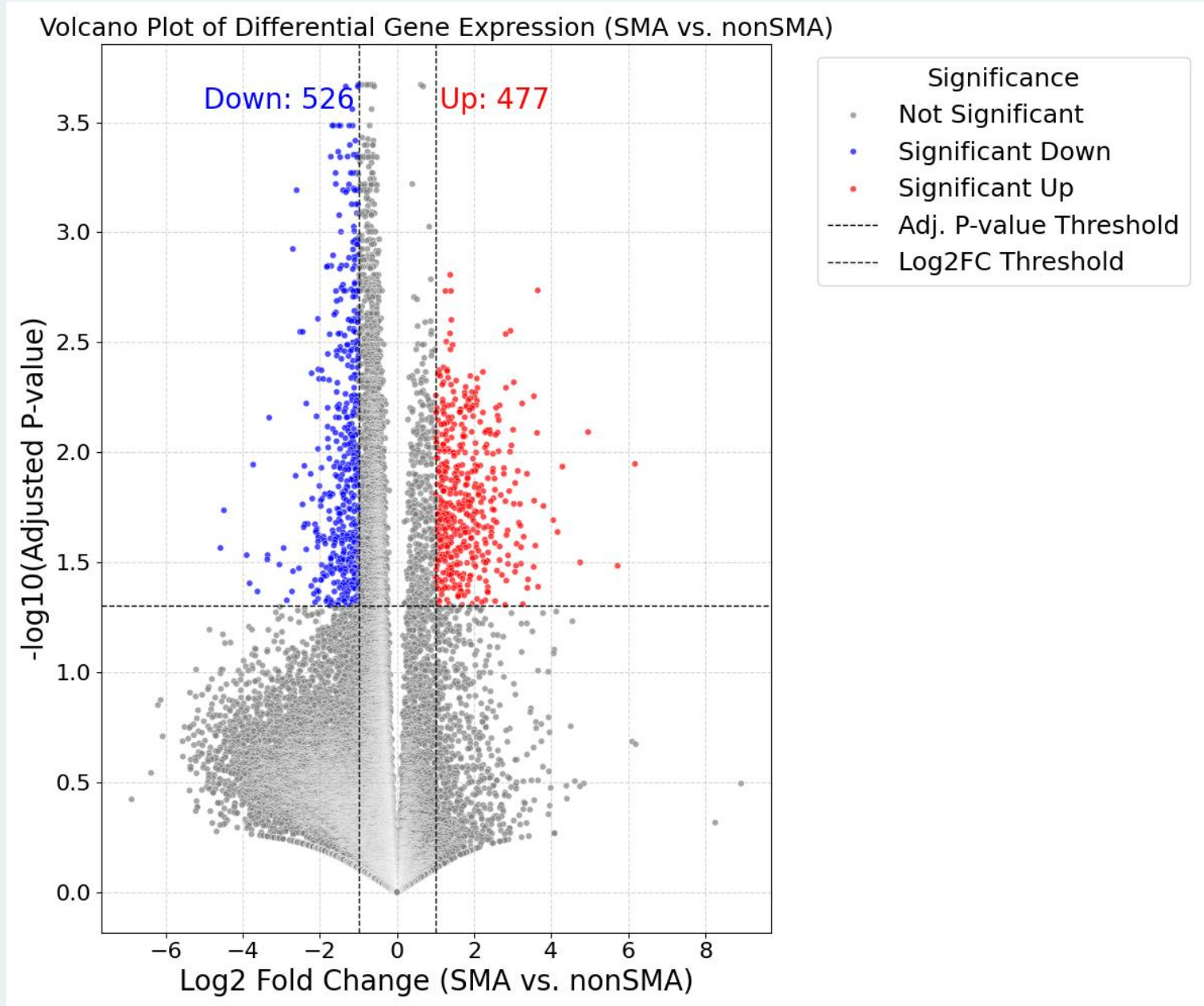
Methodology



Immune pathways behave differently in SMA vs non-SMA



Immune-related genes differ between SMA and non-SMA



Immune gene expression patterns distinguish SMA from non-SMA



Limitations and Future Work

- Our results describe differences in gene expression but lacks the mechanics of the pathways and biological effects
- We were unable to look explicitly into the data based off of factors such as age or sex.
- We could always explore more immune-related pathways.
- Compare with proteomic data

Takeaway

- SMA is associated with altered immune regulation
- Immune-related gene expression patterns and pathways differ between children with severe malarial anemia compared to children without severe malarial anemia

Thank you

Acknowledgments

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Seamus Mawe

Tucker Taft and Phyllis Yale

Vivek Philip

Citations

- Onyango, C. et al (2024). Transcriptomic and proteomic insights into host immune responses in pediatric severe malarial anemia: Dysregulation in HSP60-70-TLR2/4 signaling and altered glutamine metabolism. *Pathogens*, 13(10), 867. <https://doi.org/10.3390/pathogens13100867>
- World Health Organization. (n.d.). *World Malaria Report 2025*. World Health Organization. <https://www.who.int/teams/global-malaria-programme/reports/world-malaria-report-2025>

Thank you

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